

Customer Behavior Analysis Report

1. Project Overview

Objective

The objective of this project is to perform an end-to-end analysis of customer purchasing behavior to uncover trends, customer preferences, revenue drivers, and subscription patterns. The project demonstrates the complete analytics lifecycle, including data cleaning, exploratory data analysis, SQL-based business analysis, dashboard development, and insight generation.

Business Problem

Retail organizations generate large volumes of customer transaction data. Understanding customer purchasing behavior can help businesses:

- Improve customer retention
- Increase subscription adoption
- Optimize promotional campaigns
- Improve product strategy
- Enhance customer experience
- Increase overall revenue

Project Workflow

Dataset → Python (Cleaning & EDA) → PostgreSQL (SQL Analysis) → Power BI Dashboard → Business Recommendations

2. Dataset Summary

Dataset Overview

The dataset contains customer transaction and shopping behavior information across multiple product categories.

Key Attributes

Attribute	Description
Customer ID	Unique Customer Identifier
Age	Customer Age
Gender	Customer Gender
Item Purchased	Purchased Product
Category	Product Category
Purchase Amount	Transaction Value
Review Rating	Product Rating
Subscription Status	Customer Subscription Status
Shipping Type	Delivery Method
Discount Applied	Discount Indicator
Previous Purchases	Number of Historical Purchases
Payment Method	Mode of Payment
Frequency of Purchases	Purchase Frequency

Dataset Size

- Total Records: ~4,000
- Total Features: 18
- Data Type Mix:
 - Numerical Features
 - Categorical Features
 - Transactional Features

3. Exploratory Data Analysis (EDA) Using Python

Python libraries used:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Data Cleaning Activities

The following data quality checks were performed:

- Missing Value Analysis
- Duplicate Record Identification
- Data Type Validation
- Null Value Treatment
- Standardization of Categorical Values
- Feature Engineering

Feature Engineering

Created:

Age Group Segmentation

- Young Adult
- Adult
- Middle-aged
- Senior

Customer Segmentation

- New Customer
- Returning Customer
- Loyal Customer

Exploratory Analysis Performed

- Customer Demographic Analysis
- Revenue Distribution Analysis

- Category Performance Analysis
- Subscription Behavior Analysis
- Purchase Pattern Analysis
- Age Group Analysis
- Review Rating Analysis

Key EDA Findings

- Clothing category generated the highest sales volume.
 - Young Adult customers represented the largest customer segment.
 - Most customers were non-subscribers.
 - Purchase amounts were relatively consistent across age groups.
 - Discounts were frequently used across categories.
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4. Business Analysis Using SQL

PostgreSQL was used to answer business-driven analytical questions.

Q1. Revenue Generated by Gender

Objective:

Compare revenue contribution between male and female customers.

Insight:

Revenue contribution from both genders was analyzed to understand purchasing behavior and identify key customer groups.

Q2. Discount Users Spending Above Average

Objective:

Identify customers who used discounts but still spent above the overall average purchase amount.

Business Value:

Helps identify high-value customers who remain profitable even when promotional discounts are applied.

Q3. Top 5 Products by Average Review Rating

Objective:

Identify the highest-rated products based on customer reviews.

Business Value:

Helps product and merchandising teams prioritize popular products and improve customer satisfaction.

Q4. Average Purchase Amount by Shipping Type

Objective:

Compare spending behavior between Standard and Express shipping customers.

Business Value:

Understanding customer willingness to pay can support logistics and shipping optimization strategies.

Q5. Subscriber vs Non-Subscriber Analysis

Objective:

Compare average spending and total revenue between subscribers and non-subscribers.

Business Value:

Evaluate the effectiveness of subscription programs and determine whether subscribers represent higher-value customers.

Q6. Products with Highest Discount Adoption

Objective:

Identify products with the highest percentage of purchases involving discounts.

Business Value:

Helps marketing teams understand which products are most dependent on promotional pricing.

Q7. Customer Segmentation Analysis

Customers were segmented into:

- New Customers
- Returning Customers
- Loyal Customers

based on previous purchase history.

Business Value:

Supports targeted marketing and customer retention strategies.

Q8. Top Products Within Each Category

Objective:

Identify the three most purchased products within every product category.

Business Value:

Supports inventory planning and merchandising decisions.

Q9. Repeat Buyer Subscription Analysis

Objective:

Evaluate whether repeat customers are more likely to subscribe.

Business Value:

Provides insight into customer loyalty and subscription program effectiveness.

Q10. Revenue Contribution by Age Group

Objective:

Measure the percentage contribution of revenue by each age group.

Business Value:

Supports targeted marketing and customer acquisition strategies.

5. Power BI Dashboard

An interactive dashboard was developed to provide business users with self-service analytical capabilities.



Dashboard Components

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Review Rating

Revenue Analysis

- Revenue by Category
- Revenue by Age Group

Customer Analysis

- Subscription Distribution
- Gender Analysis
- Age Group Analysis

Product Analysis

- Sales by Category
- Product Performance Metrics

Interactive Filters

- Subscription Status
- Gender
- Category
- Shipping Type

Dashboard Benefits

- Interactive exploration of customer behavior
 - Real-time business insight generation
 - Simplified executive reporting
 - Improved decision-making capabilities
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6. Key Business Insights

Customer Behavior

- Young Adult customers generated the highest overall revenue.
- Clothing emerged as the strongest-performing product category.
- Non-subscribers represented a significant share of customers.

Product Performance

- Certain products consistently achieved higher review ratings.
- Discount adoption varied significantly across product categories.

Customer Loyalty

- Repeat buyers demonstrated stronger engagement with subscription programs.
- Loyal customers contributed a substantial share of revenue.

Revenue Trends

- Revenue was concentrated among a small number of high-performing categories.
 - Subscription customers demonstrated strong revenue-generating potential.
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7. Business Recommendations

Recommendation 1: Increase Subscription Adoption

Subscribers generally demonstrate stronger engagement and revenue contribution.

Recommended Actions:

- Introduce loyalty rewards
 - Provide subscriber-exclusive discounts
 - Launch subscription awareness campaigns
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Recommendation 2: Focus on High-Performing Categories

Clothing and top-performing categories should receive:

- Increased inventory allocation
 - Additional promotional support
 - Expanded product offerings
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Recommendation 3: Target Young Adult Customers

Young Adults represent a significant revenue-generating segment.

Recommended Actions:

- Personalized marketing campaigns
 - Digital promotions
 - Product recommendations based on purchase history
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Recommendation 4: Retain Repeat Buyers

Repeat buyers are valuable customers with high lifetime value.

Recommended Actions:

- Loyalty programs
 - Personalized offers
 - Exclusive member benefits
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Recommendation 5: Optimize Discount Strategy

Products heavily dependent on discounts should be reviewed.

Recommended Actions:

- Evaluate pricing strategies
 - Improve product positioning
 - Reduce unnecessary discount dependency
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Conclusion

This project demonstrates the complete data analytics lifecycle, including data preparation, exploratory analysis, SQL-based business problem solving, dashboard development, and actionable business recommendations.

The analysis provides meaningful insights into customer behavior, revenue drivers, product performance, and customer loyalty while showcasing practical skills in Python, SQL, PostgreSQL, and Power BI.

Technologies Used:

Python | Pandas | NumPy | PostgreSQL | SQL | Power BI | Jupyter Notebook